Learning Objective 6.2: I can decode DTMF tones from a captured audio signal.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate). The keypad grid: rows 697, 770, 852, and 941 Hz carry 1 2 3 A, 4 5 6 B, 7 8 9 C, and * 0 # D; columns are 1209, 1336, 1477, and 1633 Hz, left to right.

1. **Reading the grid.**

   (a) Which two tones does the '8' key send? The '#' key?

   '8' is row 3 and column 2: 852 Hz and 1336 Hz. '#' is row 4 and column 3: 941 Hz and 1477 Hz.

   8: 852, 1336 Hz

   #: 941, 1477 Hz

   (b) A receiver's bins show 941 Hz and 1209 Hz in one digit and 770 Hz and 1633 Hz in the next. Which two keys were pressed?

   941 and 1209 Hz is row 4, column 1: '*'. 770 and 1633 Hz is row 2, column 4: 'B'.

   *, B

   (c) In one block, the 697 Hz and 770 Hz bins are both strong and about equal, and so is the 1336 Hz bin. Which test of the L13 decoder rejects the block, and what could cause a signal like that?

   Test 2: the best row bin must beat the other three row bins by 6 dB, and here two row bins are equal. A keypad never sends two row tones at once, so the signal is something else: two keys pressed together, or speech or music with energy near both frequencies.

2. **Timing.** An auto-dialer sends each digit as 70 ms of tone followed by 30 ms of silence.

   (a) How many digits a second does it send, and how long does an 11-digit number take, from the first tone to the end of the last one?

   Each digit takes $70 + 30 = 100$ ms, so 10 digits a second.

   $$11 \times 100 - 30 = 1070 \text{ ms}$$

   rate: 10 per s

   time: 1.07 s

   (b) Does this dialer meet the standard's timing rules from Lesson 13? If not, which rule does it break, and what can go wrong at a receiver built only to the standard?

   The 70 ms tones meet the 40 ms minimum, but the 30 ms gaps are shorter than the 40 ms a receiver must handle. A receiver built only to the standard may not see a gap that short, so a repeated digit such as the two 0s in 2004 can merge into one.

3. **A shorter Goertzel block.** A designer tries Goertzel blocks of $N = 136$ samples at $f_s = 8$ kHz instead of the lesson's 205.

   (a) Find the block time and the spacing of the bins.

   $$N/f_s = 136/8000 = 17 \text{ ms}$$
   $$f_s/N = 8000/136 = 58.8 \text{ Hz}$$

   block: 17 ms

   spacing: 58.8 Hz

   (b) Find k for 852 Hz. Round it to an integer and find the frequency of that bin and its error from 852 Hz, in Hz and in percent.

   $$k = 852 \times 136/8000 = 14.48$$
   $$14 \times 58.8 = 823.5 \text{ Hz}$$

   The rounded bin is 28.5 Hz low, 3.3% of 852 Hz.

   k: 14.48

   bin: 823.5 Hz

   error: -3.3%

   (c) Explain why that rounding is a problem for a DTMF receiver, and what the L13 decoder does instead.

   The bin is centered 3.3% below the tone before the tone has any error of its own, which is almost the 3.5% at which a receiver should reject a tone, and the 852 Hz tone loses about 3.7 dB at that bin. A tone 1.5% low would sit even farther away. The L13 decoder uses $k = 14.48$ unrounded: the Goertzel recursion works for any k, so the bin sits exactly on 852 Hz.

   (d) With bins 58.8 Hz apart, is the 73 Hz spacing of 697 and 770 Hz past the first null of each bin's response? What does that mean for the 6 dB test?

   Yes. The first nulls are 58.8 Hz either side of the bin, so a tone 73 Hz away falls in the first sidelobe, about 15 dB down. The right bin still wins by much more than 6 dB, so the shorter block can separate the rows, with less margin than $N = 205$ gives.

4. **Twist.** A capture of the '9' key has a row tone of amplitude 0.20 and a column tone of amplitude 0.12.

   (a) Find the twist in dB. Does the standard require a receiver to accept it?

   $20 \log_{10}(0.12/0.20) = -4.4$ dB

   The column tone is 4.4 dB weaker, within the 8 dB a receiver must accept.

   -4.4 dB, yes

   (b) In `L13-dtmf-rx.grc`, each stock Goertzel block feeds a Complex to Mag, which shows half a tone's amplitude. What heights do the 852 Hz and 1477 Hz traces reach?

   The 852 Hz trace reaches 0.10 and the 1477 Hz trace 0.06.

   852: 0.10

   1477: 0.06

   (c) Telephone lines lose more at high frequencies, so many phones send the column tone about 2 dB above the row tone. If the line loses 6 dB more at the column tone than at the row tone, what twist arrives? Name the case the standard calls this.

   $+2 - 6 = -4$ dB: the column tone arrives 4 dB weaker than the row tone. This is the usual case, which the standard calls normal twist, and receivers accept up to 8 dB of it.

5. **The decision rules.** The L13 decoder scales each bin's power so that a lone tone gives 1, and the eight values below are one block's shares of its energy.

Bin (Hz)	697	770	852	941	1209	1336	1477	1633
Block A	0.02	0.41	0.03	0.01	0.01	0.02	0.38	0.02
Block B	0.12	0.10	0.02	0.01	0.03	0.30	0.02	0.01

(a) Apply the three tests to block A: the two-bin share, each winner's margin over the rest of its group, and the twist. Which digit does block A hold?

The best row bin is 770 Hz and the best column bin is 1477 Hz.

$$0.41 + 0.38 = 0.79 \geqslant 0.35$$
$$10 \log_{10}(0.41/0.03) = 11.4 \text{ dB} \geqslant 6$$
$$10 \log_{10}(0.38/0.02) = 12.8 \text{ dB} \geqslant 6$$
$$10 \log_{10}(0.38/0.41) = -0.3 \text{ dB}$$

The twist is inside -10 to $+6$ dB, so block A holds '6'.

(b) Which test does block B fail?

The share is $0.12 + 0.30 = 0.42$, enough, but the best row bin beats the next by only

$$10 \log_{10}(0.12/0.10) = 0.8 \text{ dB},$$

far short of 6 dB, so block B fails test 2 and holds no digit.

(c) Block A alone does not produce a digit. What else must happen before the decoder reports '6', and what does that requirement reject?

Three blocks in a row, each starting 102 samples after the last, must all hold '6'. That spans about 51 ms of signal, so short bursts of speech or noise that happen to pass the tests in one block do not count as digits.

6. **Noise.** The L13 decoder reads every number correctly down to an SNR of about 2 dB, measured across the whole 0 to 4 kHz band.

   (a) With $N = 205$, the bins are 39.0 Hz apart. About how much less noise does one bin see than the whole 4 kHz band? Give the answer in dB.

   about 20 dB

   $10 \log_{10}(4000/39.0) = 20.1$ dB

   (b) At 0 dB SNR the bins still see the tones about 20 dB above their noise, yet the decoder fails. Explain in words which test stops it.

   Test 1. At 0 dB, half of each block's energy is noise spread across the band, so the two tones can hold at most about half of it, and less in the blocks at a tone's start and end, which are only partly filled. When a block's two-bin share falls below 35%, the run of three blocks breaks and the digit is lost. The bins can still see the tones; the share test, which exists to reject speech, is what fails.